GrowCube Flashing Instructions

This procedure flashes the Arduino firmware binary to GrowCube by using `Flash_Tool.exe`.

Required Hardware

- GrowCube device
- Micro-USB OTG adapter: micro-USB male to USB-A female
- USB data cable with one end cut off
- USB-to-TTL serial adapter
- Female-to-female jumper wires
- Windows PC with `Flash_Tool.exe`

Use a real USB data cable. A charge-only cable will not work because it does not contain the data wires.

Wiring

1. Plug the micro-USB end of the OTG adapter into GrowCube.
2. Plug the USB-A end of the cut USB data cable into the OTG adapter.

Typical USB cable wire colors are:

USB cable wire	Signal	Connect to USB-to-TTL adapter
Red	Power	`5V`
Black	Ground	`GND`
White	Serial RX line	`RXD`
Green	Serial TX line	`TXD`

3. Connect the cut cable wires to the USB-to-TTL adapter with female-to-female jumper wires: red to `5V`, black to `GND`, white to `RXD`, and green to `TXD`.

Wire colors are common but not guaranteed. If the cable uses different colors, verify the cable pinout before connecting it.

Flashing

1. Plug the USB-to-TTL adapter into the PC.
2. After the wiring is connected, GrowCube should enter flashing mode.
3. Run `Flash_Tool.exe`.
4. The tool should detect the connected adapter and flash the firmware binary from the Arduino build output.
5. Wait until the tool reports that flashing is complete.

After Flashing

1. Unplug the micro-USB OTG adapter from GrowCube.
2. Disconnect the USB-to-TTL adapter from the PC.
3. Power GrowCube from the normal 12 V power connector located below the micro-USB port.

GrowCube should boot from the newly flashed firmware after normal 12 V power is connected.

If Flashing Does Not Start

Check these common causes first:

1. Incorrect wire connection. Connect the USB cable wires to the USB-to-TTL adapter as follows: red to `5V`, black to `GND`, white to `RXD`, and green to `TXD`.
2. Poor contact. Make sure every jumper wire is fully seated, the cut USB cable wires are firmly connected, and the USB-to-TTL adapter is properly plugged into the PC.